

Talk outline

1. Research Motivation and Questions
2. Data and ML Pipeline Overview
3. Exploratory Text Analysis
4. Topic Modeling (LDA, BERTopic)
5. Signal Engineering and Feature Construction
6. Classification Results and Dataset Complexity
7. Model-Agnostic Interpretability (MDI, Permutation, SHAP)
8. Conclusions and Future Directions

Motivation & Research Questions

Motivation

- ❑ Financial markets are highly sensitive to new information
- ❑ Headlines encode structured signals about macroeconomic risk, monetary policy, and geopolitical uncertainty
- ❑ Existing machine learning (ML) models often lack interpretability

Research Question

- ❑ Can we extract economically meaningful, interpretable signals from news headlines to predict market direction?

Data & ML Pipeline Overview

THE WALL STREET JOURNAL.

World Business U.S. Politics Economy Tech Markets & Finance Opinion Free Expression Arts

2026 > January > January 1

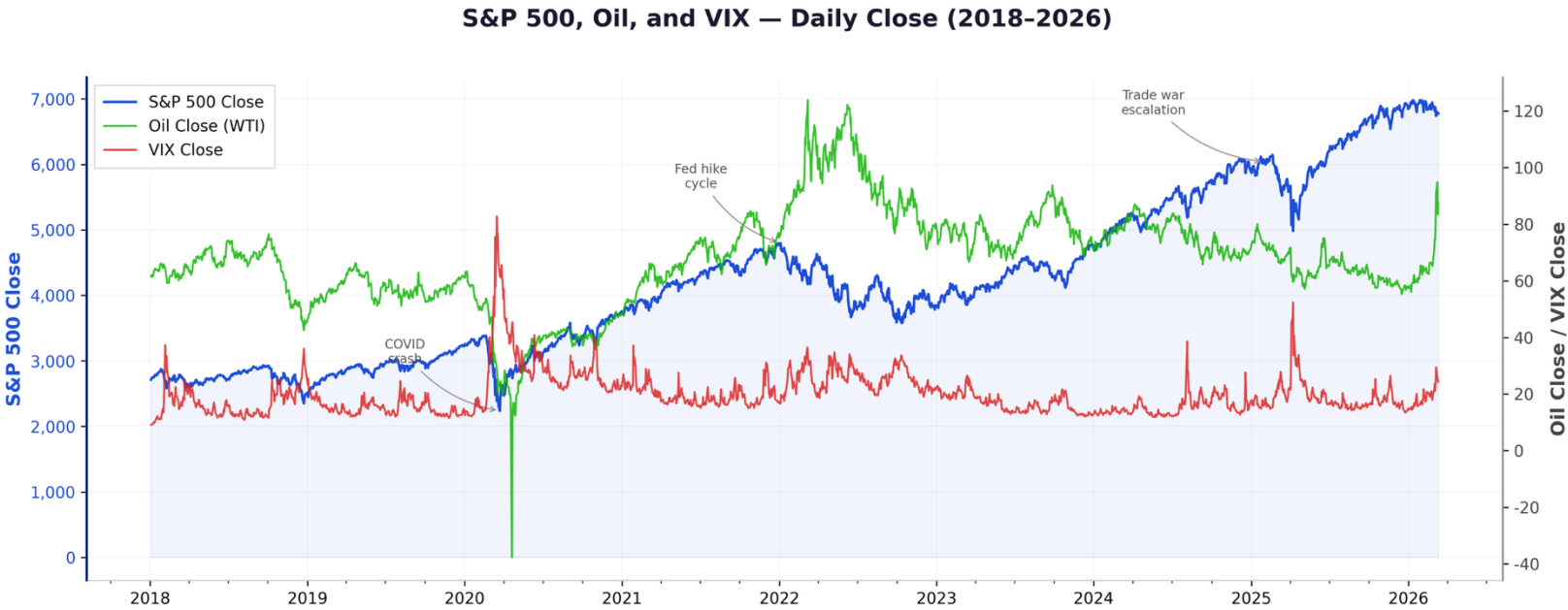
Archive for January 1, 2026

Golf, Peace Negotiations and Parties Fill Trump's Mar-a-Lago Holiday

Baidu's AI Chip Unit Kunlunxin Plans Hong Kong Listing

Modi Taps India's Consumer Power in Trade Fight With Trump

Indiana Turns the Tide on Alabama in Rose Bowl Blowout



Data & ML Pipeline Overview

WSJ Headline Corpus

- ❑ Jan 2018 – Mar 2026
- ❑ 2,992 trading days
- ❑ 8+ years of daily coverage

DOCUMENTS PROCESSED

292k

News headlines

TOKEN VOLUME

2.6M

Total words in corpus

VOCABULARY SIZE

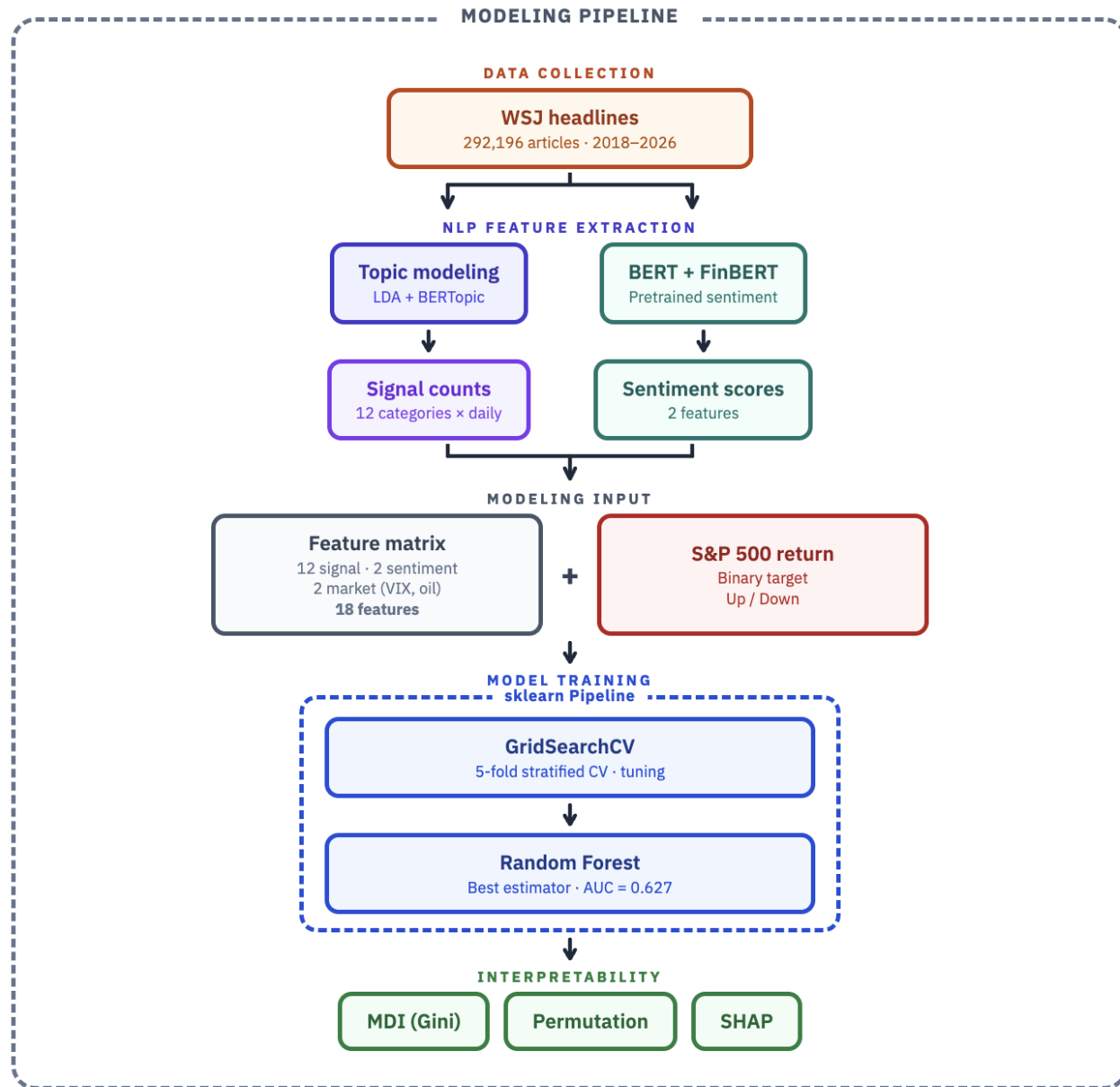
80k

Unique terms

AVG. DAILY VOLUME

~98

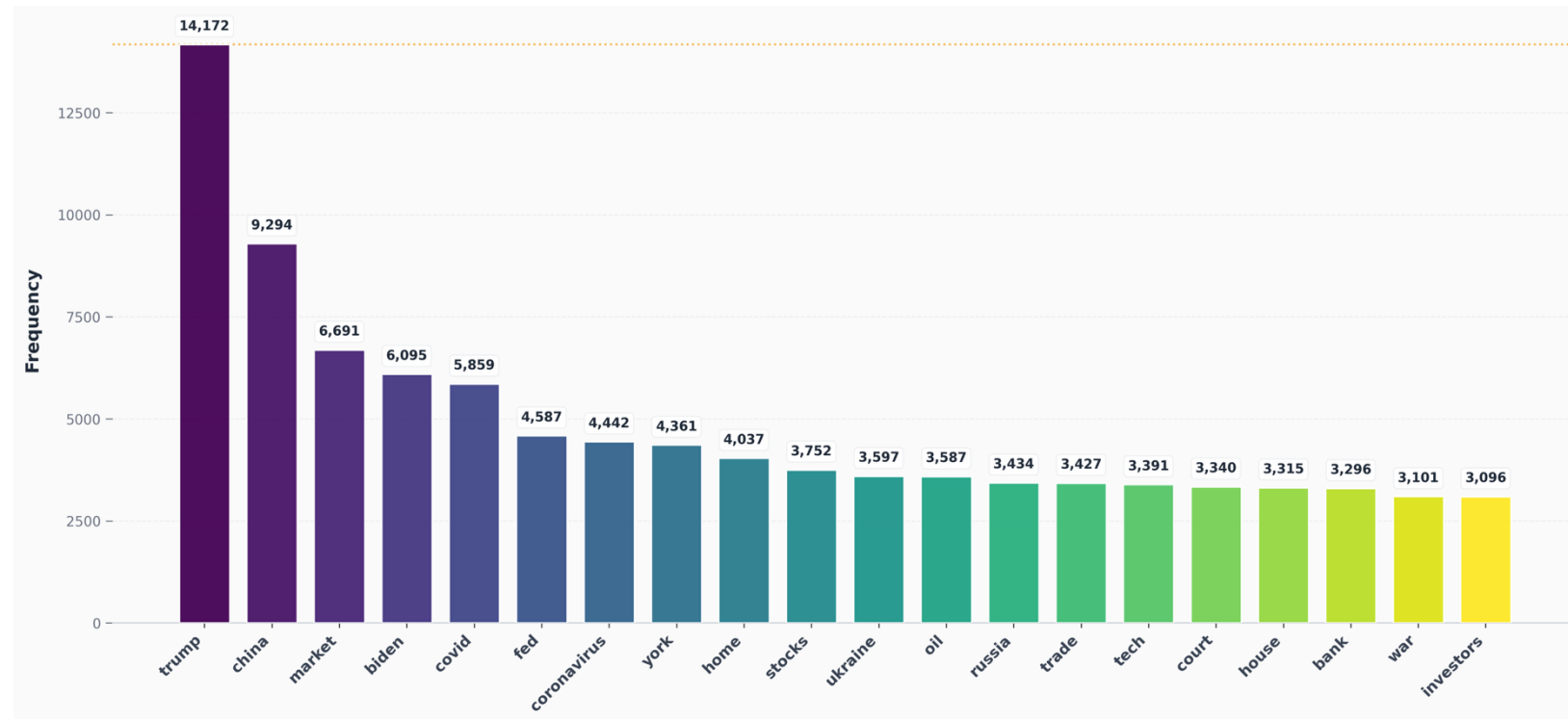
Headlines per trading day



Dominant Terms in Financial News

Exploratory Text Analysis

- ❑ Headlines are driven by politics, policy, and global events (Trump, China, Fed)
- ❑ COVID and geopolitical conflict (Ukraine, war) are highly prevalent
- ❑ Frequent terms like market, stocks, investors reflect continuous financial interpretation of news



Topic Structure in Financial Headlines

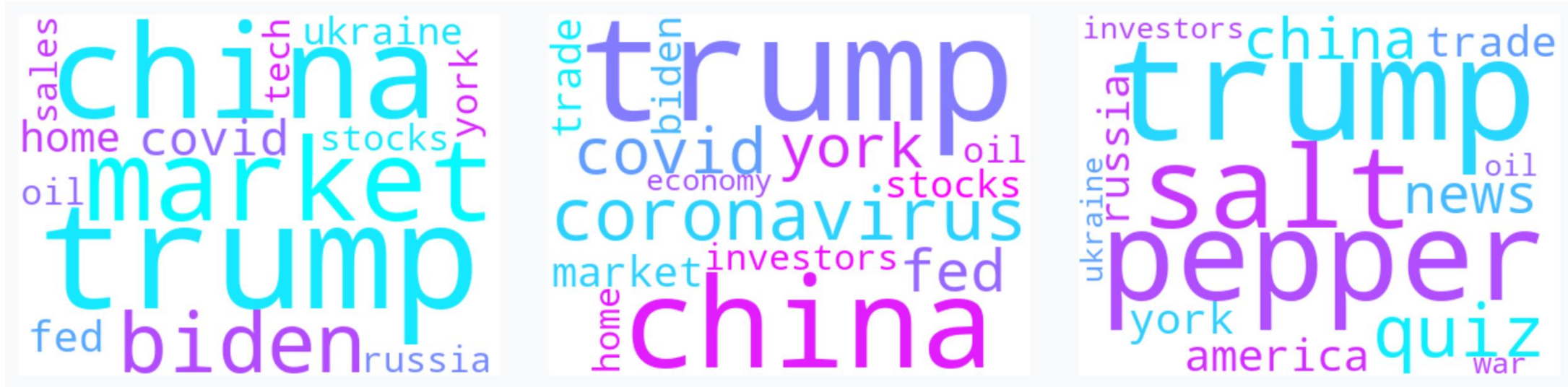
- ❑ Latent Dirichlet Allocation (LDA): Unsupervised generative model that represents each headline as a mixture of topics and each topic as a distribution over words



- ❑ Four dominant macroeconomic narratives drive investor behavior: Fed/trade policy, geopolitical conflict, pandemic & domestic politics, and commodity/tariff markets
- ❑ These latent topics directly informed the construction of keyword-based signal categories for the feature matrix

Semantic Clustering of Headlines

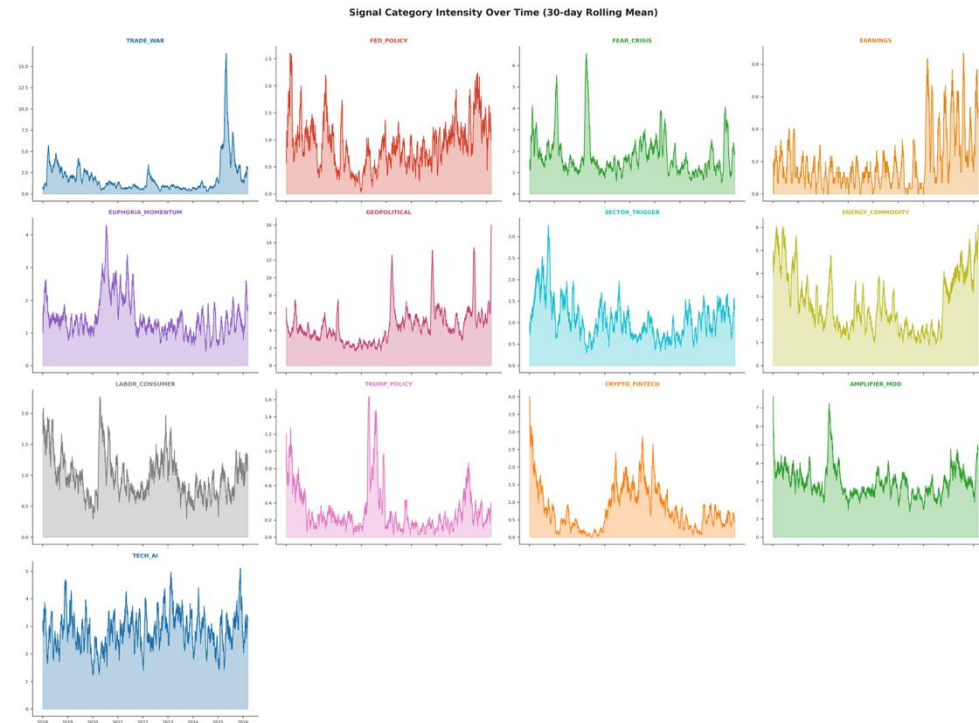
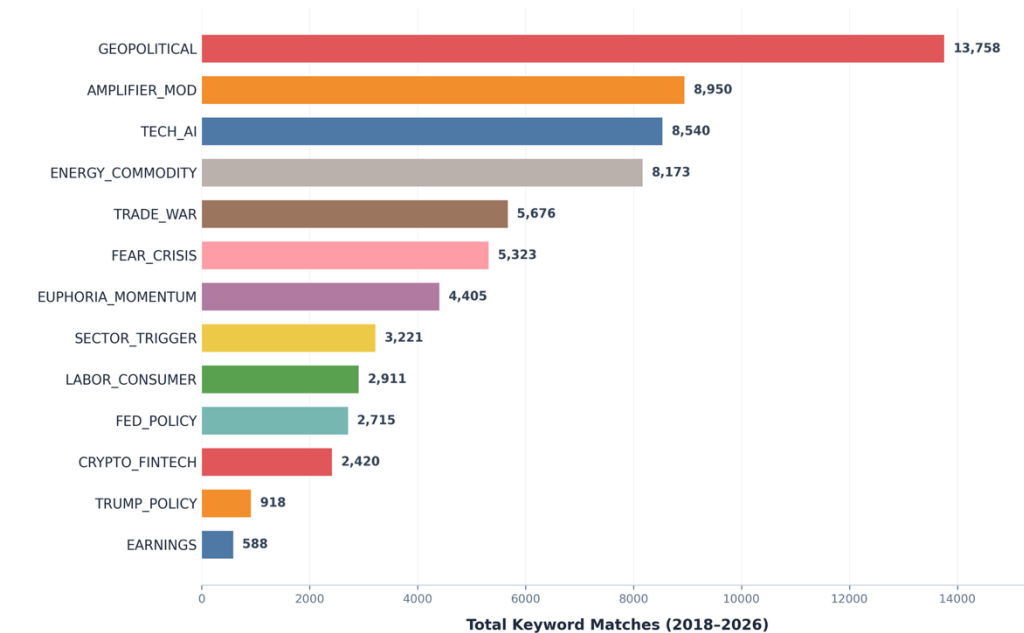
- ❑ BERTopic (BERT-based): Leverages BERT contextual embeddings to cluster headlines by semantic similarity



- ❑ Three clusters align with LDA themes — political economy and crisis/policy response with an additional noise cluster capturing non-financial lifestyle content (e.g., food, entertainment) from broader WSJ coverage

Feature Engineering: From Topic Words to Feature Matrix

- ❑ 12 keyword dictionaries derived from LDA/BERTopic and domain knowledge
- ❑ Daily keyword frequencies form a sparse signal vector capturing market attention by macro theme

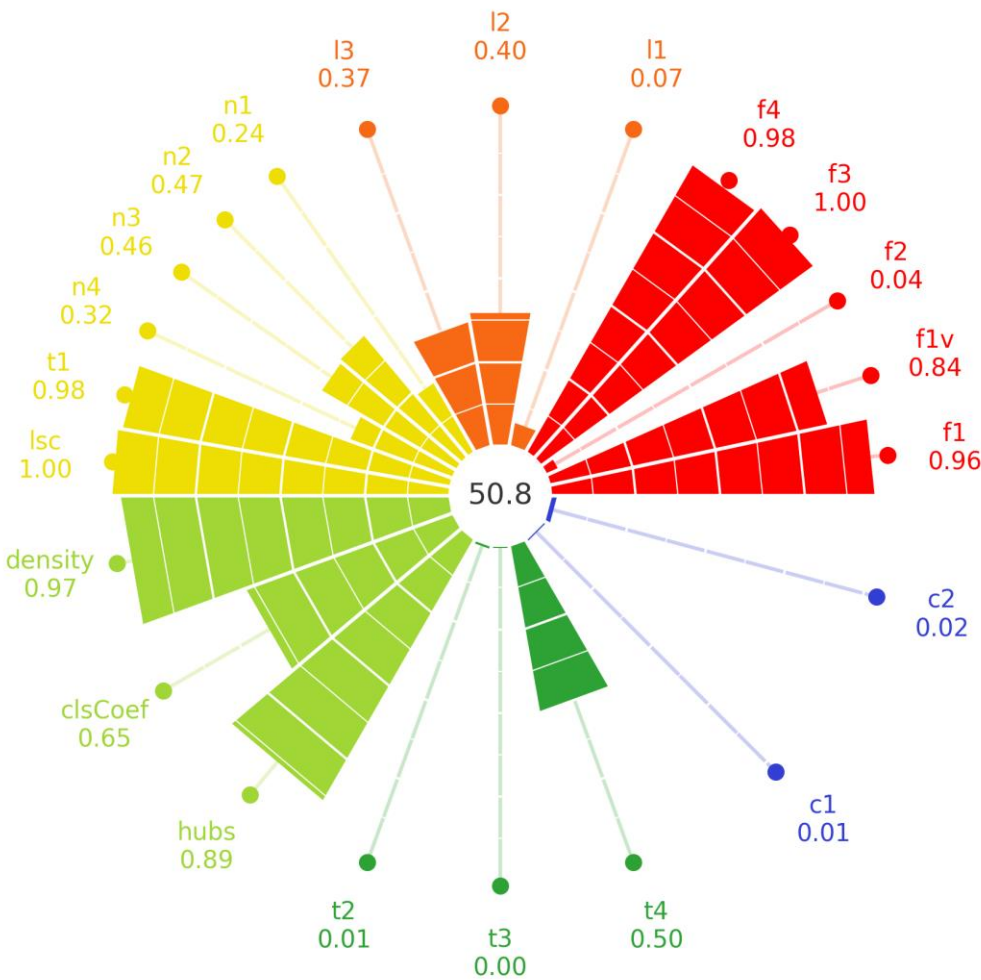
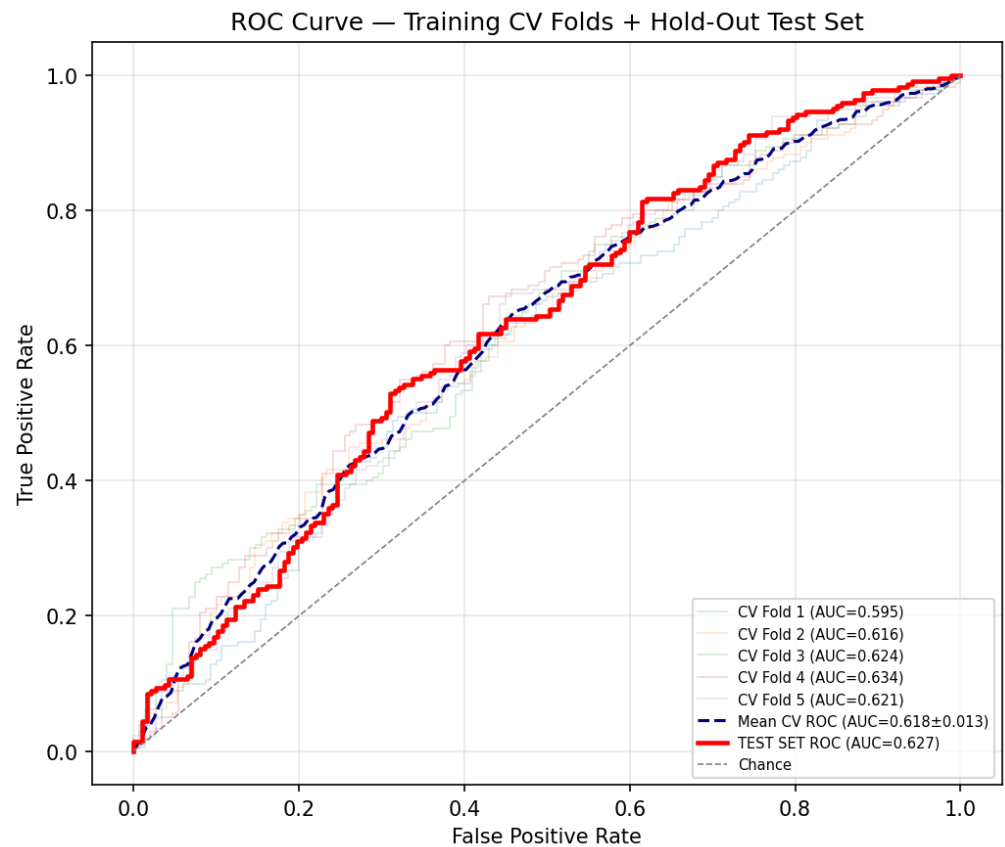


- ❑ 16 features: 12 headline signals, sentiment scores (BERT, FinBERT), and market controls (oil, VIX)



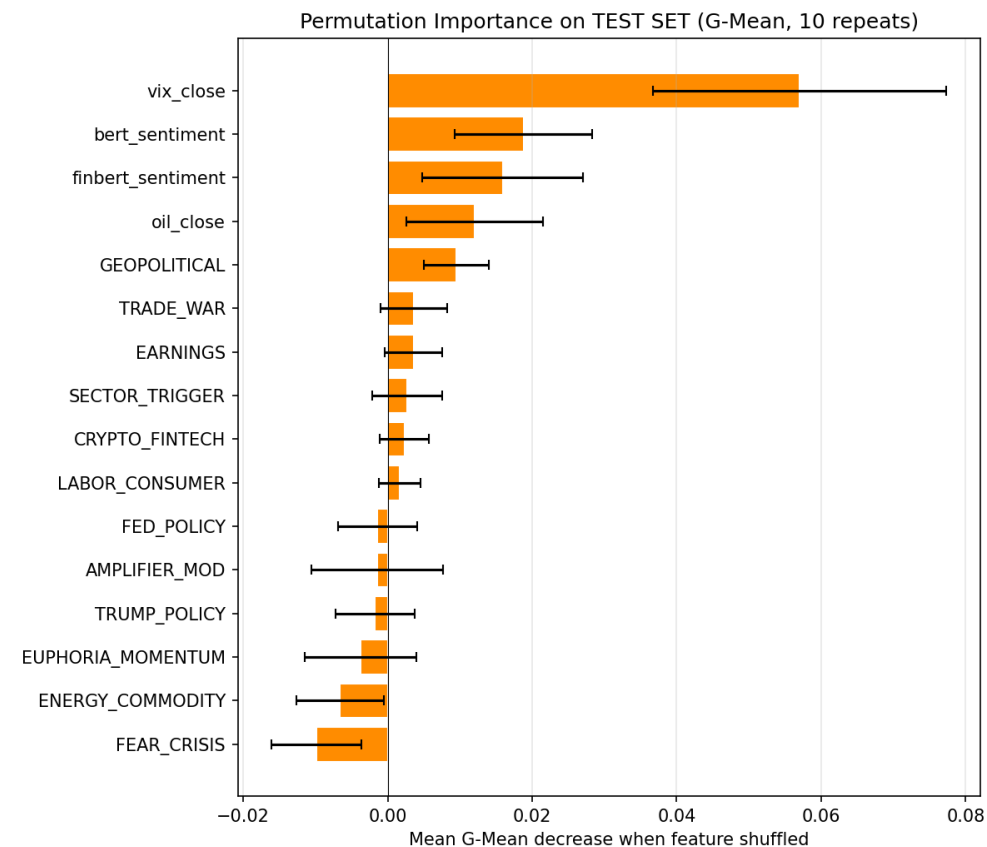
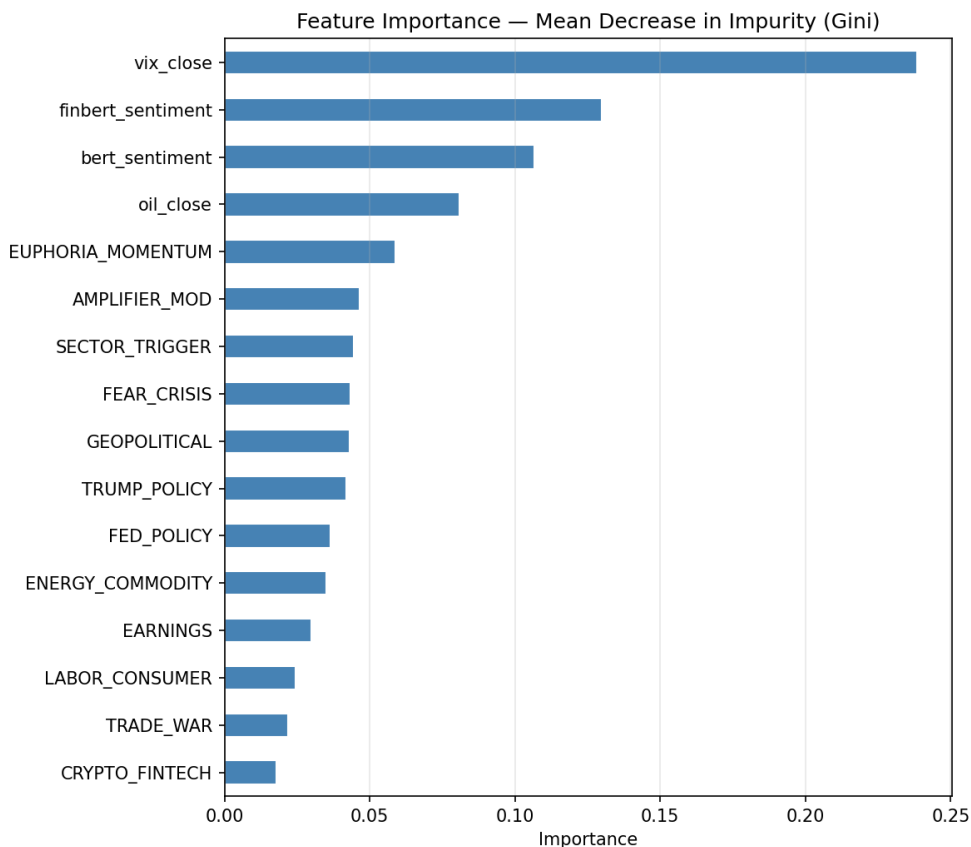
Results: Random Forest Model

☐ Stable out-of-sample performance, no overfitting



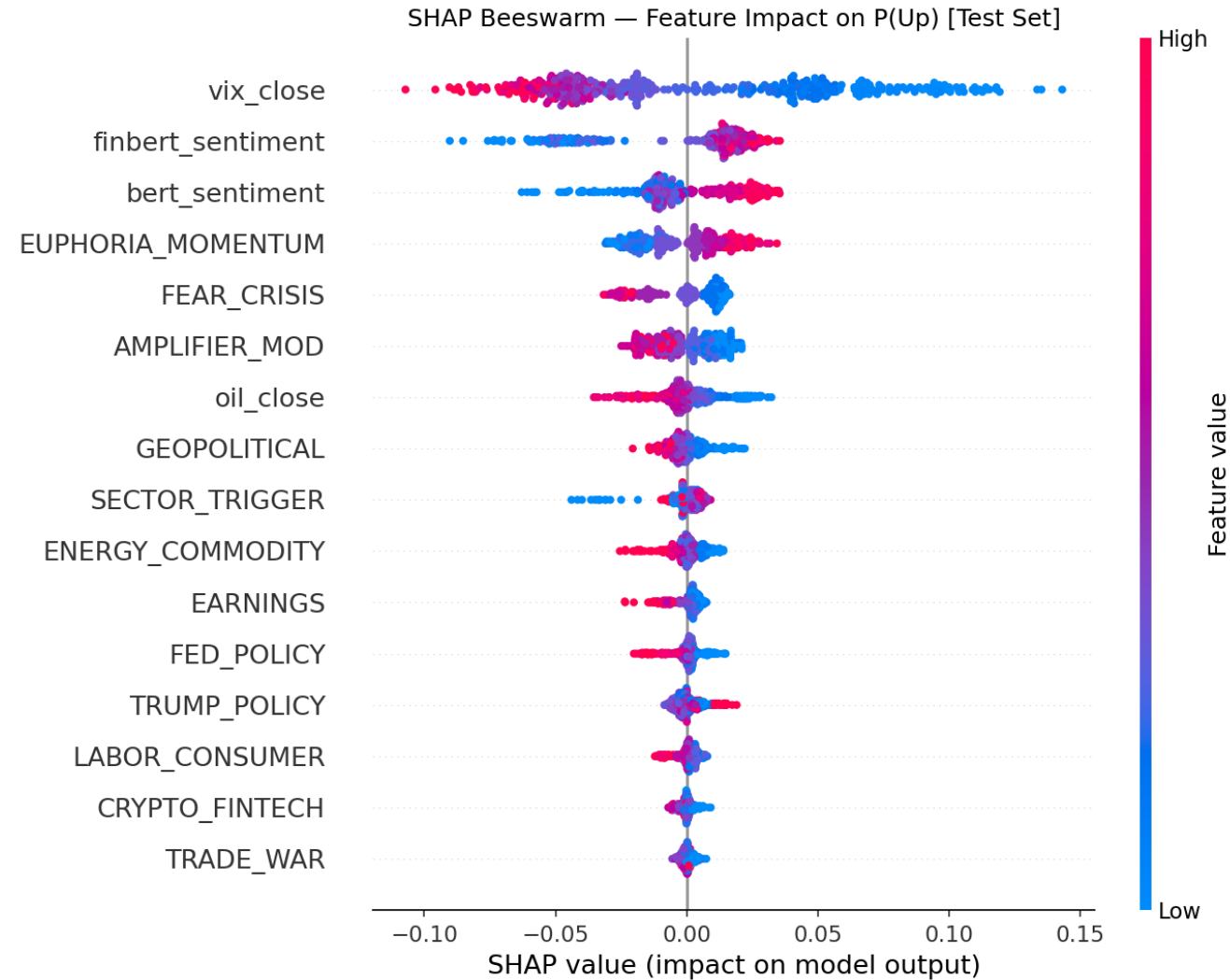
Feature Importance: MDI vs Permutation

- ❑ Market-implied volatility (VIX) dominates next-day S&P 500 direction; sentiment (BERT, FinBERT) and oil act as secondary factors
- ❑ Headline-based signals provide limited incremental explanatory power



Feature Importance: SHAP

- ❑ Market direction is primarily driven by volatility (VIX) and sentiment, while headline-based signals offer limited incremental explanatory power
- ❑ Higher FinBERT and BERT sentiment scores push predictions toward Up, confirming bullish headline tone carries measurable predictive signal for market direction



Conclusions and Future Directions

Conclusions

- ❑ LDA and BERTopic extract interpretable macroeconomic themes from 292K WSJ headlines, forming economically meaningful signal categories for direction prediction
- ❑ Market direction is driven primarily by implied volatility (VIX), commodity prices (oil), and transformer-based sentiment — consistent across MDI, permutation, and SHAP
- ❑ SHAP provides transparent, instance-level attribution of how each feature influences classification decisions

Future Directions

- ❑ Develop an agentic AI framework integrating financial statements, earnings calls, and news headlines for multi-source equity prediction